

Guideline/Protocol Title	Interim Pharmacotherapy Guidance for Multisystem Inflammatory Syndrome in Children (MIS-C)
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Committee Review	Pharmacy and Therapeutics Committee
Target Population	Pediatric patients with confirmed COVID-19 or exposure to COVID-19 within the 60 days prior to presentation
Overview	This guideline outlines the diagnostic approach to MIS-C as well as discusses the pharmacotherapy approach to managing patients with MIS-C.
Effective Date	1/14/2021
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Expiration Date	N/A
Schedule for Periodic Review	at minimum every 12 months
Implementation Strategy	Guideline will be shared on UKHC Antimicrobial Stewardship Website
Education Strategy	N/A
Primary Outcome (s)	N/A
Outcome Assessment Plan	N/A
Information Technology Needs	N/A

Interim Pharmacotherapy Guidance for Multisystem Inflammatory Syndrome in Children (MIS-C)

Revised 05/31/2023

Multisystem Inflammatory Syndrome in Children (MIS-C) has become a recognized disease process during the COVID-19 pandemic characterized by fever, inflammation and multiorgan involvement in combination with a temporal association with SARS-CoV-2. First described in April 2020, symptoms resembled that of Kawasaki Disease and Toxic Shock Syndrome. Effective January 1, 2023, the Council of State and Territorial Epidemiologists and the Center for Disease Control (CTSE/CDC) have developed and approved a new case definition. The changes included in comparison to the previous case definition are: 1) no required duration of subjective or measured fever; 2) requirement of C-reactive protein ≥ 3.0 mg/dL; 3) elimination of previous respiratory, neurologic, and renal criteria and inclusion of shock as a separate category; and 4) new requirements on timing of positive SARS-CoV-2 laboratory testing relative to MIS-C illness. Updates to the Case Definition of MIS-C are reflected herein. As this remains a diagnosis of exclusion, a multidisciplinary approach is needed for the diagnosis and management of this complication. At present, there are no FDA approved treatments for the management of MIS-C. Pharmacotherapy guidance is based on consensus guideline recommendations as well as case reports and other published evidence. Immunomodulatory dosing of IVIG with or without the addition of glucocorticoids remains the first line treatment option for patients with moderate to severe MIS-C. In patients with moderate to severe disease refractory to first line therapy, anakinra, an IL-1 antagonist, can be considered. All cases of MIS-C should be reported to the local, state, or territorial health department (**Appendix A**). As literature surrounding the management of MIS-C emerge, this document will be updated accordingly following the careful analysis.

Diagnostic Considerations for Pediatric Patients with MIS-C

Emerging reports describe a rare but serious SARS-CoV-2 associated syndrome, referred to as MIS-C. MIS-C is a diagnosis of exclusion but consists of persistent fever, elevated inflammatory markers, neutrophilia, lymphopenia, coagulopathy, and a variety of other clinical manifestations, including:

- Vasodilatory shock with normal or mildly depressed systolic function
- Cardiogenic shock with greater than or equal to moderate systolic dysfunction
- Kawasaki disease features (complete or incomplete)
- Clinical and laboratory features of cytokine storm
- Coronary artery dilation and aneurysms
- Any combination of the above

Per the Council of State and Territorial Epidemiologists and Centers for Disease Control (CTSE/CDC), the following case definition has been proposed for MIS-C. More detail can be found [here](#)

Table 1: Case Definition

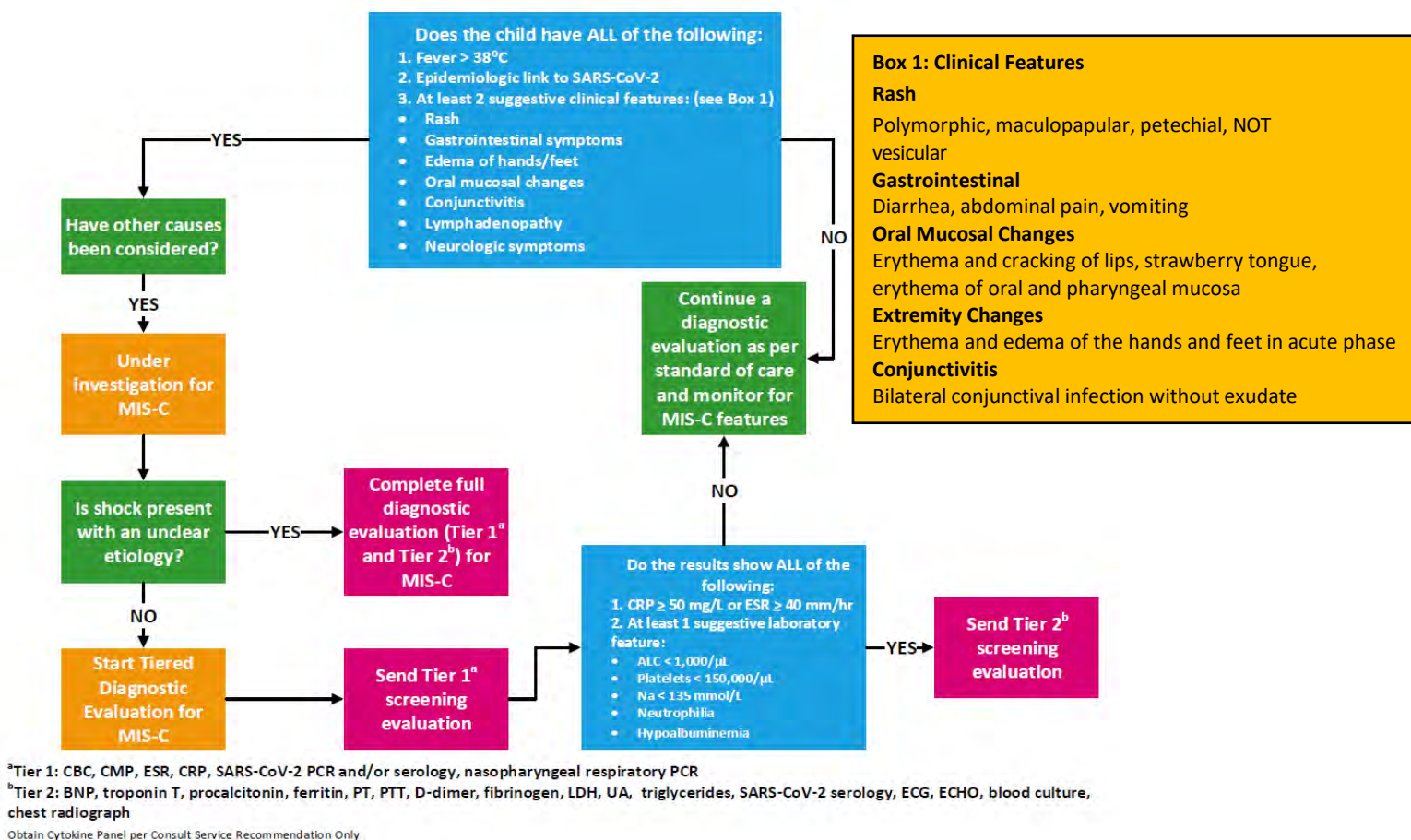
Criteria for Diagnosis	
Clinical	An illness in persons < 21 years characterized by all of the following: • Subjective or documented fever ($T \geq 38^{\circ}\text{C}$) • Clinical severity requiring hospitalization or resulting in death • Systemic inflammation indicated by C-reactive protein of ≥ 3.0 mg/dL New onset manifestations in at least two of the following (see Table 2): • Cardiac involvement • Mucocutaneous involvement • Shock • Gastrointestinal involvement • Hematologic involvement
Laboratory	Detection of COVID-19 by one of the following means: • SARS-CoV-2 RNA in a clinical specimen (up to 60 days prior to or during admission or postmortem) • SARS-CoV-2-specific antigen in a clinical specimen (up to 60 days prior to or during admission or postmortem) • SARS-CoV-2-specific antibodies in serum, plasma, or whole blood resulting in or during hospitalization
Epidemiologic	Close contact with confirmed or probably case in the 60 days prior to hospitalization • Defined as within 6 feet for at least 15 minute; also depends on exposure level and setting
Vital records	Death of a person <21 years whose death certificate lists MIS-C as an underlying cause or significantly contributing to death

Table 2: Clinical Criteria Description^a

Cardiac	Mucocutaneous	Shock	Gastrointestinal	Hematologic
• LVEF <55% • Coronary artery dilation, aneurysm, or ectasia • Elevated troponin	• Rash • Inflamed oral mucosa • Conjunctivitis • Erythema or edema of extremities	• Clinician documentation sufficient	• Abdominal pain • Vomiting • Diarrhea	• Platelets <150,000 cells/mcL • Absolute lymphocyte count of <1000 cells/mcL

^aThe presence of any one clinical presentation of a given category qualifies as a manifestation of that category

The American College of Rheumatology proposed a tiered approach to diagnosis in patients under investigation for MIS-C. The following algorithm has been proposed in the consensus guideline below (**Appendix B**)



Upon diagnosis of MIS-C, the disease process can be further classified based on severity, taking into consideration fever history, clinical features, and organ dysfunction. Suggested classification scheme is outlined below:

Disease Severity	Presentation	Admitting (or Transfer) Location
Mild	• Fever/History of Fever $\geq 38^{\circ}\text{C}$ for ≥ 1 days AND • ≥ 2 clinical features (see Box 1) • No vasopressor requirement • Minimal to no respiratory support • Minimal to no organ injury	Progressive Care Unit (Requires Telemetry)
Moderate	• Fever/History of Fever $\geq 38^{\circ}\text{C}$ for ≥ 1 day AND • ≥ 2 clinical features (see Box 1) AND • Myocardial dysfunction or fluid resistant hypotension^a OR • Significant supplemental oxygen support OR • Mild to moderate organ injury	PICU
Severe	• Fever/History of Fever $\geq 38^{\circ}\text{C}$ for ≥ 1 day AND • ≥ 2 clinical features (see Box 1) AND • Myocardial dysfunction or hypotension with vasopressor requirement OR • Non-invasive or invasive mechanical ventilation OR • Moderate to severe organ injury	PICU

^a Fluid resistant hypotension is defined as hypotension which is not responsive to two (2) 20 mL/kg isotonic fluid boluses or 40 mL/kg total

Subspecialty Consult Service Involvement:

- In all patients with suspected MIS-C, consult both Pediatric Infectious Diseases and Pediatric Rheumatology
- In all patients with confirmed MIS-C, the following subspecialty consults are recommended: Pediatric Infectious Diseases, Pediatric Rheumatology, and Pediatric Cardiology
- Roles of the involved subspecialty consults are as follows:

Subspecialty Consult Service	Role
Pediatric Infectious Disease	• Consult first to rule out alternative diagnoses ○ Empiric antibiotics are not recommended unless a serious bacterial infection is on the differential • Determine if initiation of MIS-C treatment is appropriate
Pediatric Rheumatology	• Consult first to determine if treatment for MIS-C is appropriate • Provide recommendations for pharmacotherapy needs including: IVIG, glucocorticoids, and anakinra • Provide recommendations for steroid dose/duration and taper
Pediatric Cardiology	• Consult while patient is hospitalized and MIS-C diagnosis is confirmed • Ongoing monitoring and follow-up recommendations when ECHO findings are significant for cardiac dysfunction: CAAs with Z-Score ≥ 2.5 OR LVEF $\leq 35\%$
Pediatric Hematology ^a	• Management of therapeutic anticoagulation due to DVT, PE, or coronary aneurysm • Management of prophylactic anticoagulation in select patients based on criteria below

^a Pediatric Hematology is consulted only in the setting of therapeutic anticoagulation management or if prophylactic anticoagulation is needed at discharge (**Page 8**)

Therapeutic Options for Pediatric Patients with MIS-C

This document provides guidance on the administration of pharmacotherapy for MIS-C, should it be deemed indicated by the treatment team. These recommendations are based on the American College of Rheumatology guidelines, Centers for Disease Control and Prevention, and the American Academy of Pediatrics recommendations. None of the pharmacotherapy agents listed below have FDA approval for the treatment of MIS-C.

Patients Under Investigation for MIS-C:

- Treat with supportive care
- If laboratory values are within normal limits, physical exam is normal, and patient has adequate oral intake, patient may be discharged with close PCP follow-up
- Patients with positive SARS-CoV-2 antibody does not confirm a MIS-C diagnosis

Patients with Confirmed MIS-C:

- Hospitalized patients with **CONFIRMED** SARS-CoV-2 RT-PCR, serology, or antibody or recent exposure to COVID in last 60 days along with other confirmatory guidance as outlined in the CDC case definition above (page 3) and American College of Rheumatology guideline (page 3) should be treated based on guidance from a multidisciplinary consult service
- Do not order medications until both Pediatric Infectious Disease and Rheumatology have confirmed a MIS-C diagnosis. **Pediatric Rheumatology will provide dosing recommendations for corticosteroids.**
- Treatment includes the following based on disease severity:

Disease Severity	First Line Therapy	Refractory Therapy
Mild	Methylprednisolone* (low-dose) +/- IVIG	Methylprednisolone (pulse-dose) or Anakinra
Moderate	Methylprednisolone* (low-dose) and IVIG	Methylprednisolone (pulse-dose) or Anakinra
Severe	Methylprednisolone* (pulse-dose) and IVIG	Anakinra

* When methylprednisolone is ordered in combination with IVIG, give methylprednisolone **first** followed by IVIG infusion
Additional guidance regarding dosing and laboratory monitoring in the following sections

Methylprednisolone

Indication:

- Glucocorticoids may be considered as adjunctive therapy in moderate to severe disease or those *refractory* to IVIG.
 - Diagnosis of MIS-C does not preclude the use of corticosteroids when they are otherwise indicated (i.e., asthma or catecholamine refractory shock)

Dosing:

- All initial and ongoing systemic corticosteroid dosing, including methylprednisolone, should be discussed with Pediatric Rheumatology
- When methylprednisolone is ordered in combination with IVIG, give methylprednisolone first followed by IVIG infusion

Mild or Moderate Disease	Severe Disease and/or Refractory to IVIG
Methylprednisolone ^a 2 mg/kg (maximum: 80 mg) every 24 hours followed by steroid taper over 2 to 3 weeks	Methylprednisolone ^a 30 mg/kg (maximum: 1,000 mg) every 24 hours followed by a steroid taper over 2-3 weeks

^aTransition to prednisone/prednisolone once patient is: clinically improving, hemodynamically stable (i.e., off vasopressor support) and able to tolerate oral intake

Steroid taper

- Pediatric Rheumatology to guide suggested taper duration
- Partner with primary team pharmacist to:
 - Develop a taper plan **AND** Provide education at discharge regarding taper plan

IVIG

Indication:

- IVIG should be considered a first-line therapy in patients requiring hospitalization and/or fulfill Kawasaki criteria.

Dosing:

Mild, Moderate or Severe Disease
2 g/kg ^{a,b} IV (maximum: 100 grams; round to nearest 5 gram vial)
Administration: Infuse over 10 to 12 hours. Adjust titration instructions to ensure appropriate infusion time
Pre-medications (given 30 minutes prior to IVIG) • Acetaminophen 15 mg/kg x 1 (maximum dose: 1000 mg) • Diphenhydramine 0.5 mg/kg x 1 (maximum dose: 50 mg) Caution: Due to the risk of infusion related reactions, order IVIG via the order set to ensure a hypersensitivity kit available during administration Prior to IVIG Administration: Send serum for freeze and hold for additional testing needs, including serum for Ig Profile Repeat dose is not recommended due to risk of fluid overload and hemolytic anemia associated with large doses of IVIG.

^a Dose based on actual body weight (ABW); if ABW > ideal body weight (IBW) by 20% use an adjusted body weight

Adjusted body weight: IBW + 0.4 (TBW-IBW)

> 60 inches:

Male IBW (kg) = 39 + (2.27 x [height in inches – 60])

Female IBW (kg) = 42.2 + (2.27 x [height in inches – 60])

< 60 inches:

IBW (kg) = [height (cm)]² x 1.65 / 1000

^b Cardiac function and fluid status should be assessed, may divide dose into 1 gm/kg/day x 2 days

Anakinra (IL-1 Inhibitor)

Indication:

- Hospitalized patients with moderate to severe disease that are *refractory* to IVIG and/or glucocorticoids as guided by Pediatric Rheumatology recommendations

Dosing^a:

Route	Moderate Disease	Severe Disease ^{b,c}
Subcutaneous (SC)	Anakinra 2 mg/kg SC every 12 hours (no maximum dose)	Anakinra 2 – 2.5 mg/kg/dose SC every 6 to 8 hours until inflammatory markers begin to consistently trend down
Intravenous (IV)	Anakinra 1 mg/kg/dose IV every 6 hours (no maximum dose)	Anakinra 2 – 2.5 mg/kg/dose IV every 6 hours until inflammatory markers begin to consistently trend down

^a Assess baseline renal function to determine if any dose adjustments are warranted [CrCl < 30 mL/min or ESRD: decrease frequency to every other day]

^b IV route is preferred in patients requiring > 100mg/dose, bleeding concerns, subcutaneous skin conditions, or neurologic manifestations

^c Anakinra taper recommendations will be guided by Pediatric Rheumatology, with anakinra being tapered over a minimum of 2 weeks but could approach 4 (or more) weeks pending the patient's clinical response including monitoring trend of laboratory values

Additional Pharmacotherapy Considerations:

GI Prophylaxis	• Preferred: Famotidine 0.5 mg/kg/dose PO/IV every 12 hours (every 24 hours if less than 3 months of age) • Omeprazole 1 mg/kg PO every 24 hours
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Antiplatelet and Anticoagulation Considerations in MIS-C^{1,14, 15,16}

Aspirin

Indication	Dose	Contraindications	Duration
Confirmed MIS-C	Low-Dose: 3 to 5 mg/kg/day (maximum: 81 mg)	- Platelets < 80,000/μL - Actively bleeding or at high risk for bleeding	Continued until platelet count normalizes and normal coronary arteries are confirmed at ≥ 4 weeks after diagnosis

For patients that do not meet the above criteria, antiplatelet therapy should be tailored to the patient's risk of thrombosis

Anticoagulation (The use of mechanical VTE prophylaxis is recommended in all MIS-C patients)

Indication	Agent and Dose	Therapeutic Drug Monitoring	Duration
Prophylaxis: - D-dimer ≥ 5 times the upper limit of normal OR - One or more HA-VTE risk factors (see Appendix C) HOLD anticoagulation if platelet count < 50,000/μL	Enoxaparin: < 2 months: 0.75 mg/kg SC every 12 hours ≥ 2 months of age: 0.5 mg/kg SC every 12 hours (maximum 40 mg every 12 hours) Heparin: in setting of renal dysfunction (estimated GFR < 30 mL/min/1.73 m²) > 60 kg: 5000 mg SC every 8 to 12 hours - BMI ≥ 40: 7500 mg SC every 8 hours	Routine Anti-Xa monitoring (enoxaparin only) is not required except in the following clinical situation: if renal function deteriorates (SCr > 150% of baseline) collect an Anti-Xa level 4 hours after the dose (goal anti-Xa 0.2 to < 0.5 U/mL)	- Discharge prophylactic anticoagulation maybe required, if^a: - D-dimer is ≥ 5 times the upper limit (ULN) of normal, within 24 to 36 hours of anticipated discharge AND - One or more VTE risk factors (see Appendix D) OR - Patient required ECMO^b - Discharge prophylactic anticoagulation is NOT required, if: - D-dimer is < 5 times the ULN, within 24 to 36 hours of anticipated discharge OR - D-dimer is ≥ 5 times the ULN, within 24 to 36 hours of anticipated discharge AND no VTE risk factors (see Appendix D)
Therapeutic: - CAAs present with Z-Score ≥ 10 OR absolute dimension of ≥ 8 mm (+ low dose aspirin) OR - Moderate to severe left ventricular (LV) dysfunction (LVEF $\leq 35\%$) OR - Documented thrombosis Modify anticoagulation plan based on platelet count (see Appendix E)	Enoxaparin: < 2 months: 1.5 mg/kg SC every 12 hours ≥ 2 months of age: 1 mg/kg SC every 12 hours Heparin: in setting of renal dysfunction (estimated GFR < 30 mL/min/1.73 m²) - Initiate heparin continuous infusion per institutional protocol	Anti-Xa 4 hours after the second dose (enoxaparin only) (goal anti-Xa 0.5 to 1 U/mL)	- At least 2 weeks post-discharge - Longer duration may be considered in patients with: - CAAs with Z-Score > 10 - Documented thrombosis - Ongoing moderate to severe LV dysfunction

For patients that do not meet the above criteria, anticoagulation therapy should be tailored to the patient's risk of thrombosis

See **Appendix E** for contraindications to anticoagulation therapy, including guidance based on platelet count

^a Pediatric Hematology/Oncology will review the patient and determine if based on listed criteria if patient should qualify for outpatient prophylactic anticoagulation.

^b Prophylactic anticoagulation may be considered in a patient that required ECMO after consultation with Pediatric Hematology/Oncology. The duration (at least two weeks) will be determined by ECHO results and normalization of coagulation markers at an outpatient visit.

MIS-C Laboratory and Imaging Monitoring Requirements:

- **Baseline:** All patient will have Tier 1 and Tier 2 laboratory and imaging tests completed on admission, often prior to pharmacologic intervention
 - Tier 1: CBC with Differential, CMP, ESR, CRP, SARS-CoV-2 PCR and Serology
 - Tier 2: BNP, Troponin, Procalcitonin, Ferritin, PT, PTT, D-dimer, Fibrinogen, LDH, Triglycerides, ECG, ECHO, blood culture, chest X-ray
- **Ongoing:** Frequency of laboratory testing determined by primary team, consulting services and patient's clinical status:
 - Laboratory testing needs to be considered
 - CMP, CBC, CRP, ESR, D-dimer and other Tier 1 or Tier 2 labs as clinically appropriate
 - Trending inflammatory markers aids in monitoring clinical response to initial or refractory pharmacotherapy recommendations
 - D-dimer MUST be collected within 24 to 36 hours of anticipated discharge to aid in prophylactic anticoagulation recommendations
 - Specific recommendations regarding cardiac monitoring as outlined below AND in collaboration with pediatric cardiology recommendations:
 - MIS-C Without Cardiac Involvement:
 - Laboratory Testing: trending troponin and BNP levels are not necessary unless clinical status changes/worsens
 - Other:
 - ECG: repeat is not required if initial screen is normal
 - ECHO: repeat is not required if initial screen is normal OR had minor abnormalities
 - MIS-C With Cardiac Involvement (i.e., evidence of myocarditis):
 - Laboratory Testing:
 - Troponin: repeat is not required even if initial screen is abnormal
 - BNP: repeat is not required even if initial screen is abnormal
 - Other:
 - ECG: repeat is not required if initial screen is normal
 - ECHO: repeat required while inpatient, timing is recommended by consulting cardiologist
 - Consider cardiac MRI as inpatient or within 1 to 2 months of discharge
 - MIS-C with Coronary Involvement:
 - Laboratory Testing:
 - Troponin: repeat is not required even if initial screen is abnormal
 - BNP: repeat is not required even if initial screen is abnormal
 - Other:
 - ECG: repeat is not required if initial screen is normal
 - ECHO: repeat required while inpatient, at least twice weekly, or as recommended by consulting cardiologist.
 - Consider cardiac MRI as inpatient or within 1 to 2 months of discharge

Discharge and Follow-Up:

- Patients will need appropriate follow-up with the following subspecialty services:

Primary Care Provider	• One (1) week post-discharge follow-up required (per Pediatric Rheumatology)
Pediatric Infectious Diseases	• No outpatient follow-up required
Pediatric Cardiology^a	• Two (2) to Three (3) week post-discharge follow up required • Pediatric Cardiology and Pediatric Rheumatology follow-up visits are coordinated on the same day: ○ Kentucky Clinic Location: in-person visit for both subspecialty services ○ Regional Clinic Location: ▪ Pediatric Cardiology: in-person visit ▪ Pediatric Rheumatology: telehealth visit
Pediatric Rheumatology^a	
Pediatric Hematology/Oncology^a	• In-person or telehealth visit two (2) weeks post-discharge visit required in patients discharged home on prophylactic anticoagulation (can be coordinated with Pediatric Cardiology and Pediatric Rheumatology visit) • In-person visit two (2) to three (3) week post-discharge follow-up required in patients discharged home on therapeutic anticoagulation

^aCadence of ongoing outpatient follow up is determined by the subspecialty provider

Vaccine Recommendations in Pediatric Patients following a Diagnosis of MIS-C:^{17,18,19,20,21,22}

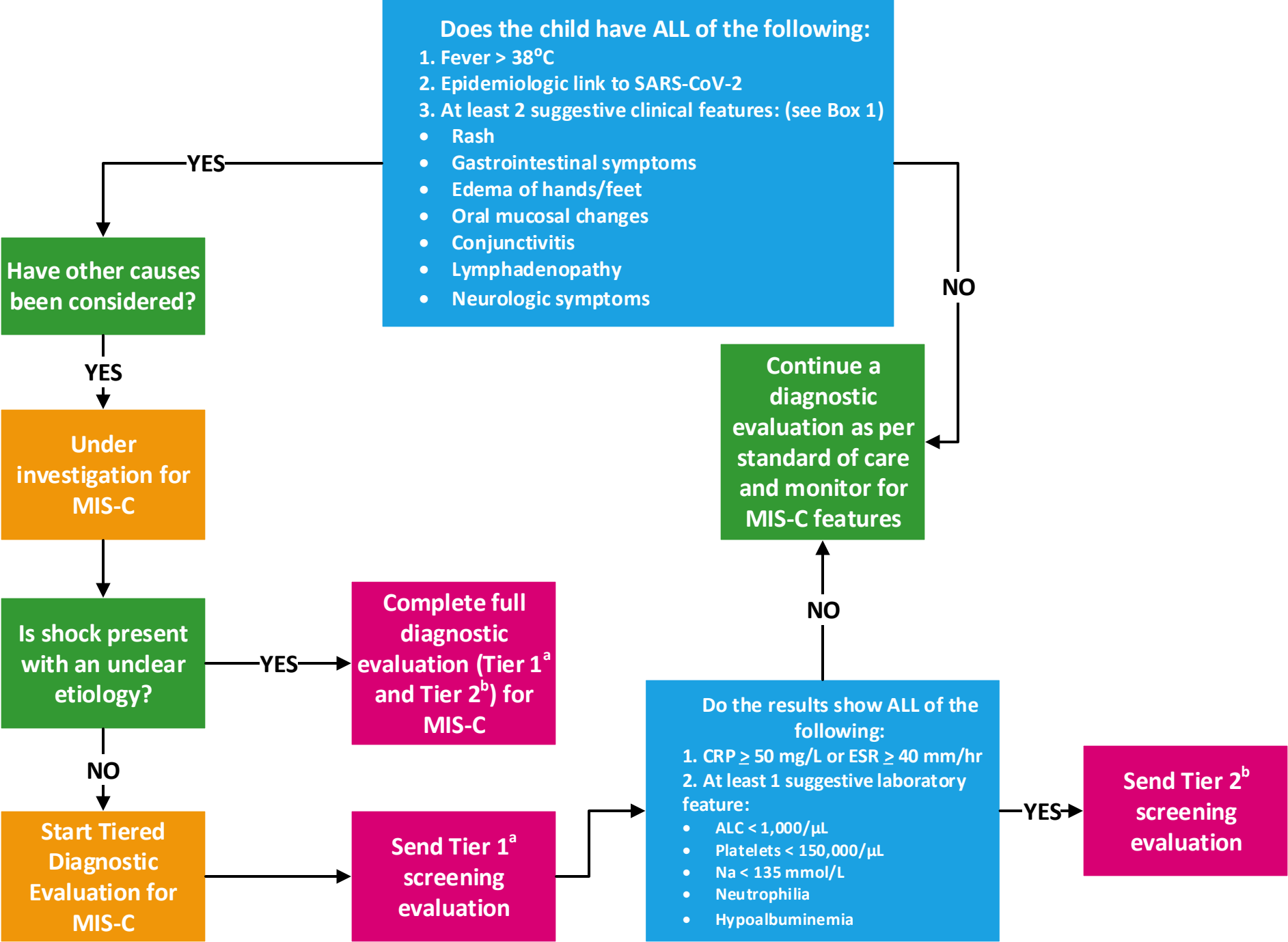
- A CDC MMWR from January 2022 reported an estimated effectiveness of 91% against MIS-C in patients 12-18 years old and received two doses of Pfizer-BioNTech. Levy M et al reported similar results in France with a hazard ratio of 0.09 (95% CI 0.04-0.21; p < 0.001) in patients that received one dose of Pfizer-BioNTech vaccine suggesting a lower incidence of MIS-C in vaccinated patients, with the risk reduced to one in one million. However, more studies need to be completed to fully elucidate the benefit of vaccines in preventing MIS-C.
- Recently published data from a multi-center cross sectional study showed no serious adverse events reported after COVID-19 vaccination in patients with a history of MIS-C, suggesting a safety profile of COVID-19 vaccination administration at least 90 days out from MIS-C diagnosis to be similar to that of the general population.
- The CDC states that the benefits of COVID-19 vaccination for people with a history of MIS-C/A (i.e., a [reduced risk of severe disease including potential recurrence of MIS-C after reinfection](#)) to outweigh a theoretical risk of an MIS-like illness or the risk of [myocarditis](#) following COVID-19 vaccination for those who meet the following two criteria:
 - o Clinical recovery has been achieved, including return to baseline cardiac function; and
 - o It has been at least 90 days after the diagnosis of MIS-C

COVID-19 Vaccine Recommendations in Pediatric Patients: The American Academy of Pediatrics recommends the COVID-19 vaccine for all children and adolescents 12 years and older who do have any contraindications.	
History of MIS-C	Consider delaying vaccination until patient has recovered from illness or 90 days from their diagnosis. However, risk versus benefit of COVID-19 reinfection needs to be weighed on an individual basis in addition to the considerations listed above.
Administration of IVIG	CDC states the administration of IVIG is unlikely to substantially impair development of protective antibody response. Therefore, no recommended length of time between IVIG and COVID-19 vaccination.

- For the most up-to-date recommendations, visit <https://www.cdc.gov/vaccines/covid-19/clinical-considerations/interim-considerations-us.html> for vaccination guidance

Appendix A: Mandatory MIS-C Reporting Guidance

- Healthcare providers should report cases of MIS-C among patients 21 years of age meeting the CSTE/CDC MIS-C surveillance case definition (described on page 3) to their local, state or territorial health department
- Clinicians can report by submitting either completed case report forms or medical records for review
 - Instructions: https://www.cdc.gov/mis/pdfs/MIS-C_case-report-form-guidance-document.pdf
 - Fillable Form: https://www.cdc.gov/mis/pdfs/MIS-C_case-report-form.pdf
- Additional information includes:
 - After-Hours Phone Number for Kentucky State Health Department:
 - After Hours/Epidemiologist On-Call Phone Number: 1-888-793-7678
 - CDC 24-Hour Emergency Operations Center: 770-488-7100
- University of Kentucky HealthCare Infection Prevention and Control (IPAC) complete the necessary documentation and stated mandated reporting of MIS-C cases



^aTier 1: CBC, CMP, ESR, CRP, SARS-CoV-2 PCR and/or serology, nasopharyngeal respiratory PCR

^bTier 2: BNP, troponin T, procalcitonin, ferritin, PT, PTT, D-dimer, fibrinogen, LDH, UA, triglycerides, SARS-CoV-2 serology, ECG, ECHO, blood culture, chest radiograph

Appendix C: Risk Factors for Hospital-Associated VTE (HA-VTE) in Children¹⁶

- Central venous catheter
- Mechanical ventilation
- Prolonged length of stay (i.e., anticipated > 3 days)
- Complete immobility (i.e., Braden Q Mobility Score = 1)
- Obesity (i.e., BMI > 95th percentile)
- Active malignancy, nephrotic syndrome, cystic fibrosis exacerbation, sickle cell disease vaso-occlusive crisis, or flare of underlying inflammatory disease (i.e., lupus, juvenile idiopathic arthritis, inflammatory bowel disease)
- Congenital or acquired cardiac disease with venous stasis or impaired venous return
- Previous history of VTE
- First-degree family history of VTE before age 40 years or unprovoked VTE
- Known thrombophilia (i.e., protein S, protein C, or antithrombin deficiency; factor V Leiden; factor II G20210A; persistent antiphospholipid antibodies)
- Pubertal, post-pubertal or age > 12 years
- Receiving estrogen-containing oral contraceptive
- Status-post splenectomy for underlying hemoglobinopathy

Appendix D: Risk Factors for VTE Prophylaxis at Discharge in Children with MIS-C¹⁶

- Central venous catheter
- Complete immobility (i.e., Braden Q Mobility Score = 1)
- Obesity (i.e., BMI > 95th percentile)
- Active malignancy, nephrotic syndrome, cystic fibrosis exacerbation, sickle cell disease vaso-occlusive crisis, or flare of underlying inflammatory disease (i.e., lupus, juvenile idiopathic arthritis, inflammatory bowel disease)
- Congenital or acquired cardiac disease with venous stasis or impaired venous return
- Previous history of VTE
- First-degree family history of VTE before age 40 years or unprovoked VTE
- Known thrombophilia (i.e., protein S, protein C, or antithrombin deficiency; factor V Leiden; factor II G20210A; persistent antiphospholipid antibodies)
- Receiving estrogen-containing oral contraceptive
- Status-post splenectomy for underlying hemoglobinopathy

Appendix E: Contraindications to Pharmacologic Anticoagulation in Children

- Active or potential bleeding
- Coagulopathy/severe liver disease
- Heparin induced thrombocytopenia (HIT)
- Platelet count:
 - **HOLD** prophylactic anticoagulation if platelet count is < 50,000
 - **MODIFY** therapeutic anticoagulation based on the following criteria:
 - CONTINUE treatment dosing for platelet count $\geq$ 50,000
 - SWITCH to prophylactic dosing for platelet count between 20,000 and < 50,000
 - HOLD anticoagulation for platelet count < 20,000
- Imminent/urgent surgical procedure
- Lumbar puncture, spinal injection or epidural catheter removal within past 2 hours
- Multiple trauma with high bleeding risk

Appendix F: MIS-C Order Set in Epic

Pediatric Multisystem Inflammatory Syndrome in Children Order Panel

Can be found through various search features including MIS-C, COVID, multisystem, etc.

Name	User Version Name	Type
PED - MULTISYSTEM INFLAMMATORY SYNDROME IN CHI...		Order Panel

Currently **three** panels to select from: Evaluation, Treatment and Refractory

PED - Multisystem Inflammatory Syndrome in Children (MIS-C) ✓ Accept

- [Information for Healthcare Providers about Multisystem Inflammatory Syndrome in Children \(MIS-C\)](#)
- [UKHC Enterprise MIS-C Protocol](#)

☐ MIS-C Evaluation

☐ MIS-C Treatment

☐ MIS-C Refractory Treatment

Next Required ✓ Accept

Check here for information on MIS-C (from the CDC) and a link to the UKHC MIS-C Treatment Guideline!

Within the **three** panels you will find the needed information as it relates to diagnostics and treatment of a patient under investigation for or diagnosed with MIS-C

MIS-C Evaluation: Tier 1

- All of the following are pre-selected

☒ MIS-C Evaluation

☒ Tier 1 Evaluation

☒ CBC and Differential

Once, today at 1007, For 1 occurrence
Release to patient in MyChart: Immediate
Blood, Blood, Venous

☒ Comprehensive Metabolic Panel, Plasma

Once, today at 1007, For 1 occurrence
Release to patient in MyChart: Immediate
Blood, Blood, Venous

☒ Sedimentation Rate, Automated

Once, today at 1007, For 1 occurrence
Release to patient in MyChart: Immediate
Blood, Blood, Venous

☒ C-Reactive Protein, Plasma

Once, today at 1007, For 1 occurrence
Release to patient in MyChart: Immediate
Blood, Blood, Venous

☒ SARS CoV-2/COVID-19 by PCR - Rapid

Once, today at 1007, For 1 occurrence
Release to patient in MyChart: Immediate
Swab, Nasopharynx

☒ Nasopharyngeal Respiratory Panel

Once, today at 1007, For 1 occurrence
Release to patient in MyChart: Immediate
Swab, Nasopharynx

☒ Freeze and Hold

Once, today at 1007, For 1 occurrence
Release to patient in MyChart: Immediate
Blood, Blood, Venous

☒ ECG Pediatric

Once, today at 1007, For 1 occurrence
ECG Indications: Other
Drug/Chemical Overdose? No
Pacemaker? No

☐ Blood Culture (Aerobic/Anaerobes Set)

Once, Starting 8/25/22, Blood, Blood, Venous

☐ XR Chest 1 View

Routine, One time imaging, Starting 8/25/22

☐ Inpatient consult to Peds Infectious Disease

☐ Inpatient consult to Peds Rheumatology

For patients under investigation for MIS-C, start with a **Tier 1 Evaluation**.

When selecting "MIS-C Evaluation", **Tier 1 Evaluation** is defaulted. If needed, be sure to select Blood Culture AND Chest Radiograph

For patients with an abnormal **Tier 1 Evaluation** select the **Tier 2 Evaluation** (since **Tier 1 Evaluation** is always defaulted, be sure to uncheck **Tier 1 Evaluation** those orders if already collected!)

For patients presenting with a very high suspicion for MIS-C, select the **Tier 2 Evaluation** (in combination with the defaulted **Tier 1 Evaluation**).

MIS-C Evaluation: Tier 2

- All of the following are pre-selected

☒ MIS-C Evaluation

☐ Tier 1 Evaluation

☒ Tier 2 Evaluation

☒ N-Terminal Probnp, Plasma

Once, today at 1010, For 1 occurrence
Release to patient in MyChart: Immediate
Blood, Blood, Venous

☒ Troponin T, High Sensitivity, 0 Hour Plasma

Once, today at 1010, For 1 occurrence
Release to patient in MyChart: Immediate
Blood, Blood, Venous

☒ Procalcitonin, Plasma

Once, today at 1010, For 1 occurrence
Release to patient in MyChart: Immediate
Blood, Blood, Venous

☒ Ferritin, Serum

Once, today at 1010, For 1 occurrence
Release to patient in MyChart: Immediate
Blood, Blood, Venous

☒ Prothrombin Time/INR

Once, today at 1010, For 1 occurrence
Release to patient in MyChart: Immediate
Blood, Blood, Venous

☒ APTT

Once, today at 1010, For 1 occurrence
Release to patient in MyChart: Immediate
Blood, Blood, Venous

☒ D DIMER, QUANTITATIVE

Once, today at 1010, For 1 occurrence
Release to patient in MyChart: Immediate
Blood, Blood, Venous

☒ Fibrinogen, Quantitative (Clottable)

Once, today at 1010, For 1 occurrence
Release to patient in MyChart: Immediate
Blood, Blood, Venous

☒ Lactate Dehydrogenase, Plasma

Once, today at 1010, For 1 occurrence
Release to patient in MyChart: Immediate
Blood, Blood, Venous

☒ Triglycerides, Plasma

Once, today at 1010, For 1 occurrence
Release to patient in MyChart: Immediate
Blood, Blood, Venous

☒ Urinalysis with reflex microscopic

Once, today at 1010, For 1 occurrence
Release to patient in MyChart: Immediate
Urine, Urine, Clean Catch

For patients undergoing a **Tier 2 Evaluation**, Pediatric Infectious Diseases and Pediatric Rheumatology are defaulted consult services!

☒ Inpatient consult to Peds Infectious Disease

Reason for Consult? MIS-C
 Patient Care Team: PID PEDS INFECTIOUS DISEASE
Consulting Privilege: Recommendations and ordering privileges
Did you contact the consultant? Yes

☒ Inpatient consult to Peds Rheumatology

Reason for Consult? MIS-C
 Patient Care Team: PRH PEDS RHEUMATOLOGY
Consulting Privilege: Recommendations and ordering privileges
Did you contact the consultant? Yes

☒ Echo, Pediatric Transthoracic (TTE) Complete

Routine, Once, today at 1010, For 1 occurrence
Where should test be performed? Bedside
Reason for Exam: Multisystem Inflammatory Syndrome in Children, MIS-C

☐ Echo, Pediatric Transthoracic (TTE) Limited

Routine, Once, Starting 8/25/22 Reason for Exam: Multisystem Inflammatory Syndrome in Children, MIS-C

MIS-C Treatment

- All of the following are pre-selected

☒ MIS-C Treatment

- Information for Healthcare Providers about Multisystem Inflammatory Syndrome in Children (MIS-C)

☒ Inpatient consult to Peds Cardiology

Reason for Consult? MIS-C

Patient Care Team: PCD PEDS CARDIOLOGY

Consulting Privilege: Recommendations and ordering privileges

Did you contact the consultant? Yes

☒ PEDIATRIC IVIG PREMEDS

☒ acetaminophen (Tylenol) 160 MG/5ML solution 640 mg

640 mg (rounded from 650 mg), Oral, Once, 1 dose, today at 1030, Routine

 The original dose of 1,408 mg (15 mg/kg Once) exceeded the recommended single dose limit of 650 mg. The dose has been automatically changed to 650 mg.

☒ diphenhydramine (BENADRYL) 12.5 MG/5ML elixir 47.5 mg

47.5 mg (0.5 mg/kg × 95 kg), Oral, Once, 1 dose, today at 1030, Routine

☐ ondansetron (Zofran) oral liquid

0.15 mg/kg, Oral, Once

☐ methylPREDNISolone (SOLU-Medrol) IV

2 mg/kg, Intravenous, Once

☒ immune globulin (human) (Gammagard) infusion 100 g

100 g, Intravenous, Once, 1 dose, today at 1030, Routine

 The original dose of 190 g (2 g/kg Once) exceeded the recommended single dose limit of 100 g. The dose has been automatically changed to 100 g.

☐ ASA - aspirin chewable tablet

4 mg/kg (Dosing Weight), Oral, Daily, Starting 8/25/22

☐ PULSE DOSE - methylPREDNISolone sod suc (PF) (SOLU-Medrol) injection

30 mg/kg/day (Dosing Weight), Intravenous, Every 24 hours, Starting 8/25/22, Routine

☐ LOW DOSE - methylPREDNISolone sod suc (PF) (SOLU-Medrol) injection

2 mg/kg/day (Dosing Weight), Intravenous, Every 24 hours, Starting 8/25/22, Routine

☐ IF D-Dimer >5x NORMAL OR HA-VTE (+) Consider enoxaparin (Lovenox) injection

0.5 mg/kg (Dosing Weight), Subcutaneous, 2 times daily, Starting 8/25/22, Routine

MIS-C Refractory Treatment

☒ MIS-C Refractory Treatment

☐ MODERATE IV - anakinra (Kineret) injection

1 mg/kg (Dosing Weight), Intravenous, Every 6 hours, Starting 8/25/22, Routine

☐ MODERATE SUBQ - anakinra (Kineret) injection

2 mg/kg, Subcutaneous, Every 12 hours, Starting 8/25/22, Routine

☐ SEVERE IV - anakinra (Kineret) injection

2 mg/kg, Intravenous, Every 6 hours, Starting 8/25/22, Routine

☐ SEVERE SUBQ - anakinra (Kineret) injection

2 mg/kg, Subcutaneous, Every 6 hours, Starting 8/25/22, Routine

For patients that have refractory disease, the order set can be utilized to capture use of anakinra. The MIS-C Refractory Treatment order panel includes dose, route, and frequency recommendations for anakinra in the setting of moderate or severe MIS-C. Just likely with methylprednisolone dosing, Pediatric Rheumatology will provide recommendations on dosing strategy.

For questions or clarifications regarding the MIS-C Order Set, please reach out to a member of the MIS-C Task Force.

The most up to date guideline recommendations can be found [here!](#)

Be sure to always check the UKHC COVID-19 Website for additional information (<https://covid-19.ukhc.org/documents/>).

Upon confirmation of a diagnosis of MIS-C, this order set can be utilized to aid in treatment decisions. For those patients with mild, moderate or severe MIS-C, treatment is often IVIG **and** methylprednisolone. In the MIS-C Treatment order panel, IVIG is defaulted, including the premedication needs. Methylprednisolone dosing is driven by Pediatric Rheumatology – low dose vs. pulse dose methylprednisolone – be sure to select the appropriate dose based on recommendations. Additional supportive medications can be found within the MIS-C Treatment order panel: aspirin and enoxaparin.

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Consensus Guidelines:

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